POLICY GRADIENT LAB

REINFORCE · value baselines · GAE

A reproducible study of variance reduction and sample efficiency

24 completed runs / 4 environments / primary training seed: 1

CartPole-v0

Best evaluation-batch return: 200.0

Configuration: cartpole_lb_na · training seed 1

HalfCheetah-v4

Best evaluation-batch return: 610.4

Configuration: cheetah_baseline_na · training seed 1

LunarLander-v2

Best evaluation-batch return: 236.4

Configuration: lunar_lander_lambda0.99 · training seed 1

InvertedPendulum-v4

Best evaluation-batch return: 1,000.0

Configuration: pendulum · training seed 1

Peaks are descriptive single-batch observations, not confidence estimates.
Final returns, trailing means, raw logs and limitations appear inside.

Based on Berkeley CS285 Spring 2026 Homework 2
Generated directly from versioned experiment CSVs.

Measured evaluation returns (seeds listed per run)

Experiment | peak | final | last 10 mean | first target steps

cartpole_lb_na (seed 1) | 200.0 | 200.0 | 200.0 | 44274
cartpole_lb_rtg_na (seed 1) | 200.0 | 200.0 | 200.0 | 69020
cartpole_lb_rtg (seed 1) | 200.0 | 200.0 | 200.0 | 73115
cartpole_lb (seed 1) | 200.0 | 200.0 | 165.0 | 77340
cartpole_na (seed 1) | 200.0 | 200.0 | 200.0 | 21151
cartpole_rtg_na (seed 1) | 200.0 | 200.0 | 165.5 | 18004
cartpole_rtg (seed 1) | 200.0 | 200.0 | 186.7 | 41159
cartpole (seed 1) | 200.0 | 122.0 | 126.3 | 20711
cheetah_baseline_bgs1 (seed 1) | -54.8 | -54.8 | -154.3 | None
cheetah_baseline_na (seed 1) | 610.4 | 370.9 | 503.3 | 385000
cheetah_baseline (seed 1) | 389.2 | 104.9 | 127.0 | 435000
cheetah_baseline (seed 2) | 263.8 | 208.1 | 208.2 | None
cheetah_baseline (seed 3) | 99.0 | -60.6 | -11.5 | None
cheetah_baseline_torch (seed 1) | 386.6 | 374.2 | 324.3 | 475000
cheetah (seed 1) | -204.9 | -231.2 | -235.2 | None
pendulum (seed 1) | 1000.0 | 6.3 | 71.7 | 431837
pendulum_tuned_fast (seed 1) | 1000.0 | 142.9 | 138.9 | 26939
pendulum_tuned (seed 1) | 1000.0 | 222.4 | 372.7 | 40028
pendulum_tuned_small (seed 1) | 1000.0 | 1000.0 | 932.3 | 36372
lunar_lander_lambda0.95 (seed 1) | 93.9 | -19.9 | -40.3 | None
lunar_lander_lambda0.98 (seed 1) | 169.2 | 68.4 | 50.4 | 268736
lunar_lander_lambda0.99 (seed 1) | 236.4 | 227.6 | 99.2 | 241257
lunar_lander_lambda0 (seed 1) | 62.5 | -97.3 | -64.0 | None
lunar_lander_lambda1 (seed 1) | 180.2 | 51.6 | 101.5 | 315719

1. CartPole: returns, normalization and batch size

Small batch (1,000). Last-ten mean evaluation returns: trajectory 126.3, reward-to-go 186.7, normalized trajectory 200.0, normalized reward-to-go 165.5. Without normalization, reward-to-go performed better by this endpoint metric.

Normalization changed the trajectory last-ten evaluation mean by +73.7 points.

Normalization changed the reward-to-go last-ten evaluation mean by -21.2 points.

Large batch (4,000). Last-ten mean evaluation returns: trajectory 165.0, reward-to-go 200.0, normalized trajectory 200.0, normalized reward-to-go 200.0. Without normalization, reward-to-go performed better by this endpoint metric.

Normalization changed the trajectory last-ten evaluation mean by +35.0 points.

Normalization changed the reward-to-go last-ten evaluation mean by +0.0 points.

With normalized reward-to-go, increasing the batch changed the last-ten mean from 165.5 to 200.0. The first 200-return evaluations occurred at 18004 and 69020 training steps respectively (None means never reached).

Reward-to-go is generally preferred because an action cannot influence past rewards. Removing those past terms preserves the expected score-function gradient while usually reducing its variance. Relative discounting weights future rewards from each decision.

Normalization rescales the update and subtracts a batch mean; it is a useful heuristic but introduces finite-batch bias. Compare normalized and unnormalized rows above rather than assuming it always helps. Batch size trades gradient noise against interaction cost: 100 iterations at 4,000 steps use roughly four times the data of 1,000-step batches. The charts compare actual training steps, not iteration counts. CartPole plots show training-batch average return, as in the assignment; the comparison table uses evaluation return.

2. HalfCheetah: learned baseline

cheetah: peak -204.9; final -231.2.

cheetah_baseline: peak 389.2; final 104.9. Baseline MSE (mean over optimizer steps on each training batch) changed from 178.8 to 22.2.

cheetah_baseline_bgs1: peak -54.8; final -54.8. Baseline MSE (mean over optimizer steps on each training batch) changed from 193.6 to 30.3.

Adding a baseline changed the last-ten evaluation mean from -235.2 to 127.0. Reducing its updates to one changed that mean to -154.3; final training-batch baseline MSE was 22.2 with five updates and 30.3 with one.

Baseline seed repeats

The seed-1 baseline reached the peak target but missed the final-return target. Two additional seeds use exactly the same learning hyperparameters. All attempts are retained.

Seed 1: peak 389.2, final 104.9, last-ten mean 127.0.

Seed 2: peak 263.8, final 208.1, last-ten mean 208.2.

Seed 3: peak 99.0, final -60.6, last-ten mean -11.5.

Optional advantage normalization

The optional normalized-baseline run reached peak 610.4, final 370.9, and last-ten mean 503.3. It adds only advantage normalization to the seed-1 baseline configuration. This additional setting is reported separately from the prescribed unnormalized ablation.

Original-sampler reference

A further seed-1 baseline trial uses the original PyTorch action sampler, with all learning hyperparameters unchanged: peak 386.6, final 374.2, last-ten mean 324.3. The NumPy and PyTorch backends have equivalent policy distributions but different seeded action streams; this diagnostic checks whether low final scores persist with the reference sampler. It is labeled separately from the three NumPy seed trials.

Unnormalized final-return target: met in the recorded trials; best final evaluation 374.2 (cheetah_baseline_torch, seed 1). The normalized optional result is kept separate from this check.

The one-gradient-step ablation keeps the baseline learning rate at 0.01 and reduces value updates from five to one. This slows fitting to the current batch; policy outcomes above show whether that change helped this seed. Value targets change with the policy, so training MSE need not decrease monotonically, and lower training MSE alone does not prove better control. The baseline uses raw targets.

3. LunarLander: generalized advantage estimation

Best peak: lunar_lander_lambda0.99, 236.4. Best trailing-ten mean: lunar_lander_lambda1, 101.5.

At λ=0 the final evaluation was -97.3; at λ=1 it was 51.6. These endpoint measurements and the intermediate settings above describe how the chosen horizon affected this particular training seed. The comparison is descriptive; each setting needs more training seeds for a reliable ranking.

λ=0 uses one-step TD residuals, reducing variance while depending heavily on value accuracy. λ=1 telescopes to Monte Carlo reward-to-go minus the baseline under the zero-tail boundary convention, reducing bootstrap bias while increasing variance. Intermediate λ mixes horizons. The observed ranking above and curves below reflect this seed, not a universal monotonic relationship. All five runs retain the PDF’s fixed batch size, learning rate, architecture, discount, and iteration budget.

The official starter uses standard LunarLander-v2 without an additional action-noise wrapper; this protocol is preserved. Headless runs explicitly prune visual-only exhaust particles after each original environment step. Observations, rewards and termination were bitwise identical in 5,000 comparison steps; learning hyperparameters are unchanged. See PROVENANCE.md and the recorded cleanup flag.

4. InvertedPendulum: sample efficiency

Earliest observed 1,000-return evaluation: pendulum_tuned_fast, at 26,939 training steps. The below-100K target was met.

json
{
  "batch_size": 1000,
  "discount": 0.99,
  "learning_rate": 0.02,

```
  "n_layers": 2,
  "layer_size": 64,
  "use_reward_to_go": true,
  "normalize_advantages": true,
  "use_baseline": true,
  "baseline_gradient_steps": 10,
  "gae_lambda": 0.95
}
```

The earliest-crossing run ended at 142.9. The highest trailing-ten mean belonged to pendulum_tuned_small (932.3), whose final evaluation was 1000.0. First threshold crossing and sustained performance are different criteria.

The matched 1,000-step-batch candidates differ only in actor learning rate: 0.01 first reached 1,000 at 40028 steps and ended at 222.4; 0.02 first reached it at 26939 steps and ended at 142.9. The larger rate accelerated the initial crossing in this seed, while both candidates later lost performance.

Tuning reduces batches from 5,000 to 500 or 1,000 transitions to allow more frequent policy updates per interaction budget, and combines reward-to-go, normalization, a value baseline, γ=0.99 and GAE. The candidates also vary learning rate and λ. These coupled changes are a configuration search, not a controlled causal attribution of gains to any one hyperparameter. All attempts are retained.

Reproduction and limitations

Exact command lines are in commands.sh; per-run flags and source revision are beside each CSV. The eight CartPole, three HalfCheetah, five LunarLander, and four InvertedPendulum configurations, plus two additional HalfCheetah baseline seeds, the optional normalized-baseline run, and a reference PyTorch-sampler baseline, are generated by scripts/experiments.py.

Single-seed comparisons and peak selection can be optimistic. A larger study should repeat every setting over independent seeds and evaluate saved policies over many held-out episodes. Training steps exclude evaluation interactions as required by the assignment. Time-limit tails are zero, not bootstrapped. Videos show individual rollouts of final checkpoints and do not establish average performance.

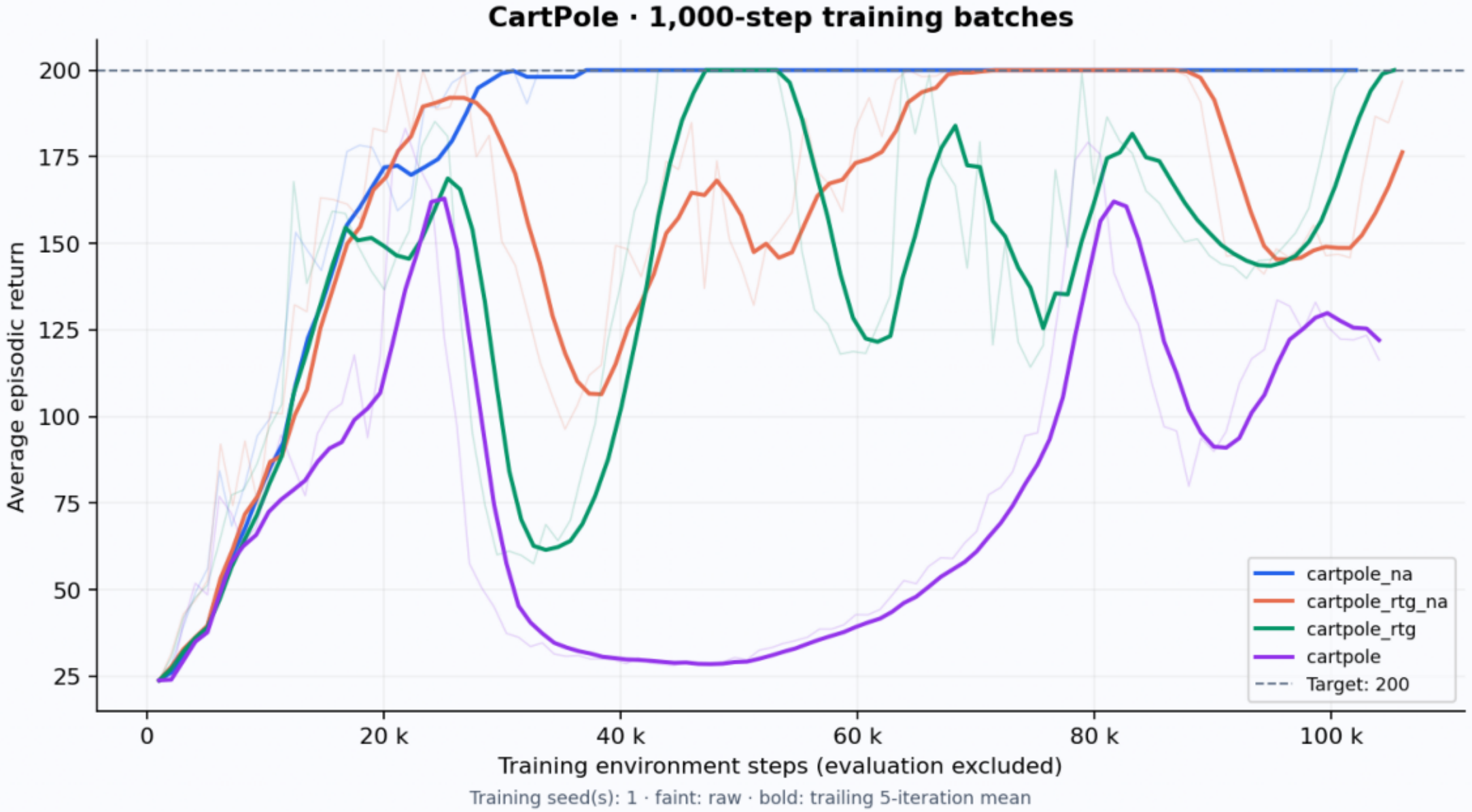
Fresh-episode evaluation of selected final checkpoints

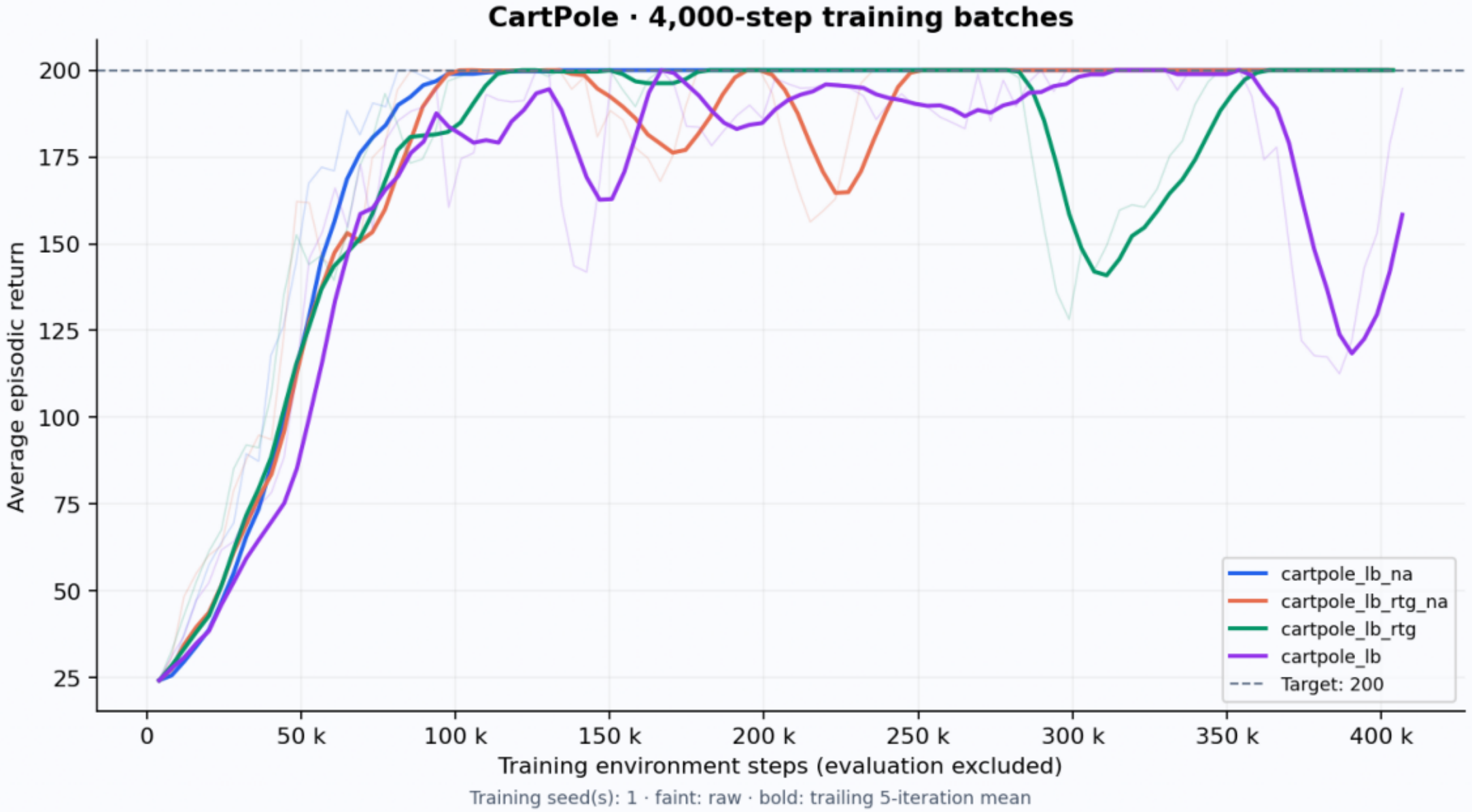
Configurations are selected from the benchmark comparisons. Each frozen final policy is then evaluated stochastically on 30 new, individually seeded episodes. These bootstrap intervals measure episode variation for one trained policy, not training-seed uncertainty.

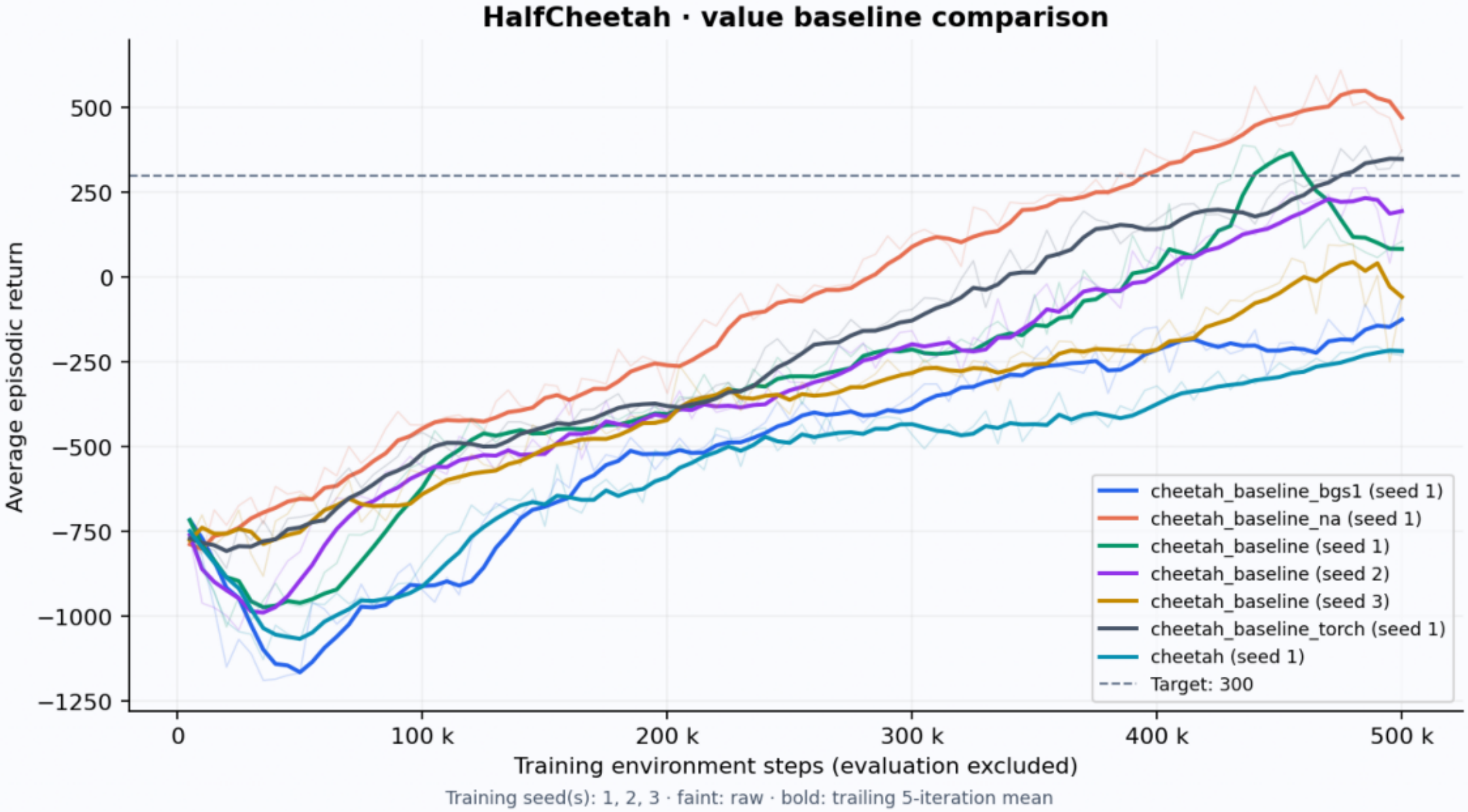
Selected policy	Episodes	Mean return	Standard deviation	Bootstrap 95% CI
[cartpole_na](evaluation/cartpole.json)	30	200.0	0.0	[200.0, 200.0]
[cheetah_baseline_na](evaluation/cheetah.json)	30	354.5	71.8	[329.2, 379.9]
[cheetah_baseline_torch](evaluation/cheetah_reference.json)	30	409.0	55.9	[388.4, 428.3]
[cheetah_baseline](evaluation/cheetah_seed1.json)	30	96.1	111.4	[52.3, 131.6]
[lunar_lander_lambda0.99](evaluation/lunar.json)	30	120.0	143.7	[66.3, 167.4]
[pendulum_tuned_small](evaluation/pendulum.json)	30	949.8	199.2	[871.1, 1000.0]

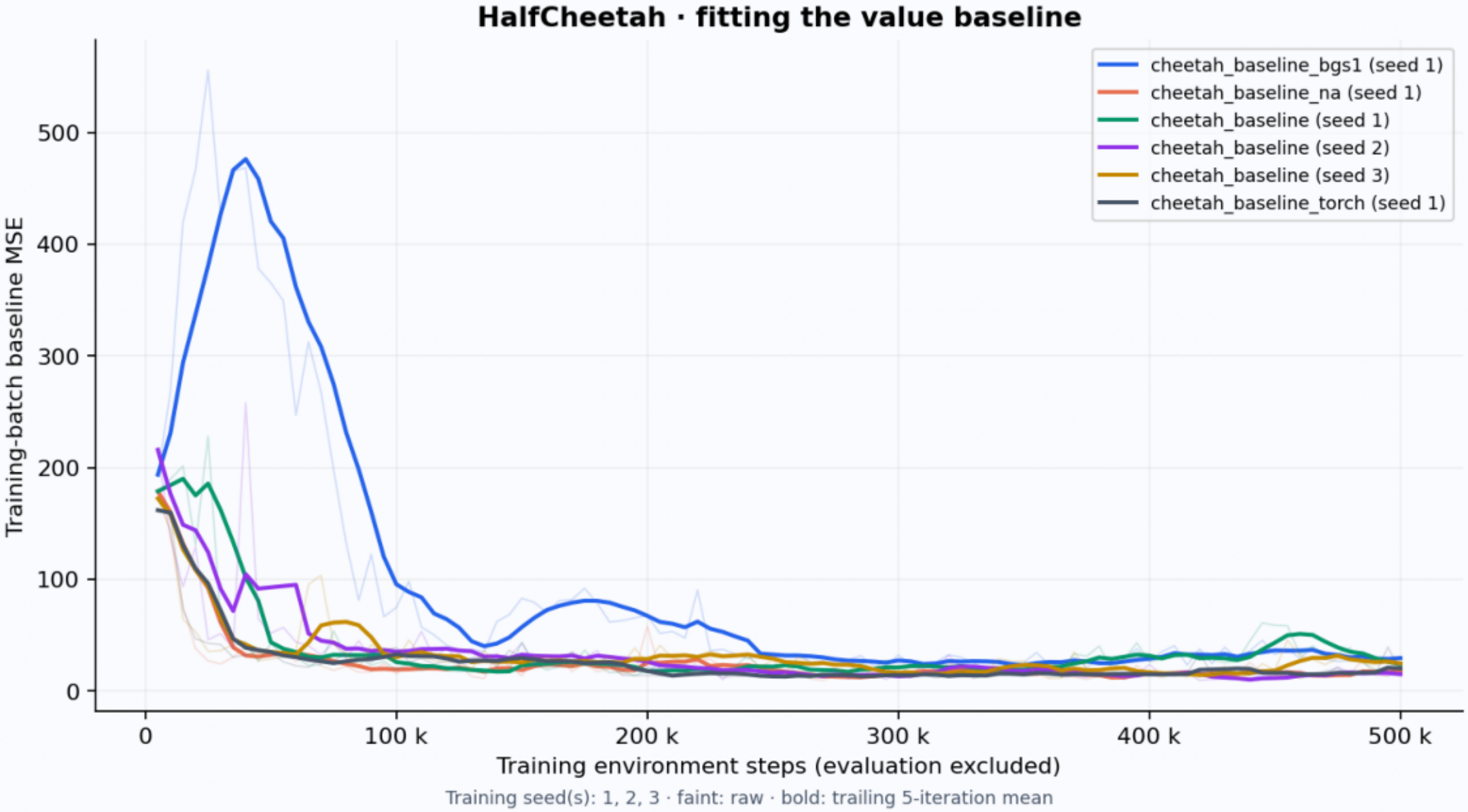
Demonstration selection

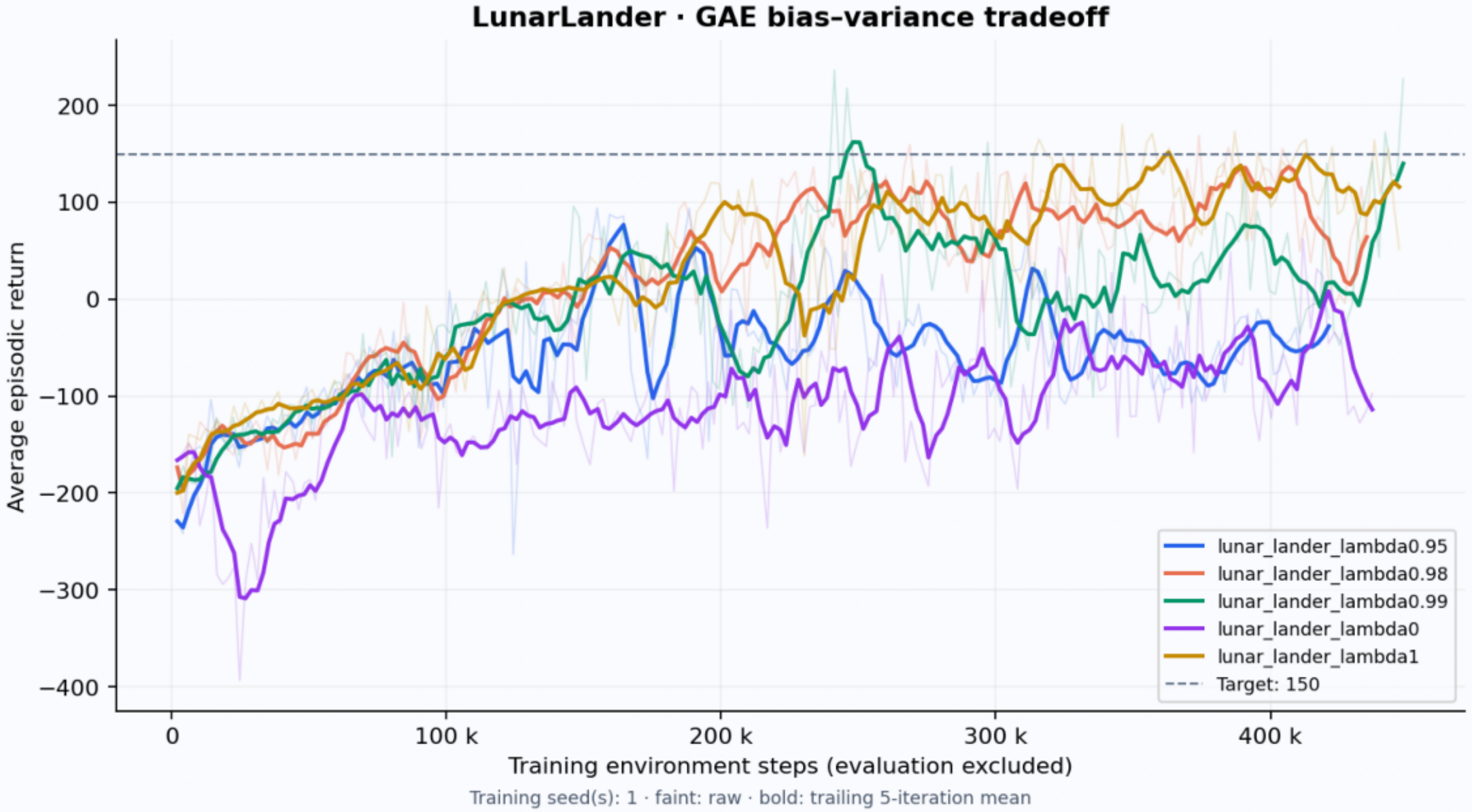
GIFs are short previews; MP4s retain full episodes. The LunarLander showcase is an explicitly selected successful example, not a representative average. The selection log records every candidate tried, and the original seed-42 failure video is retained. The fresh-episode table above includes all 30 benchmark outcomes independently of demo selection.

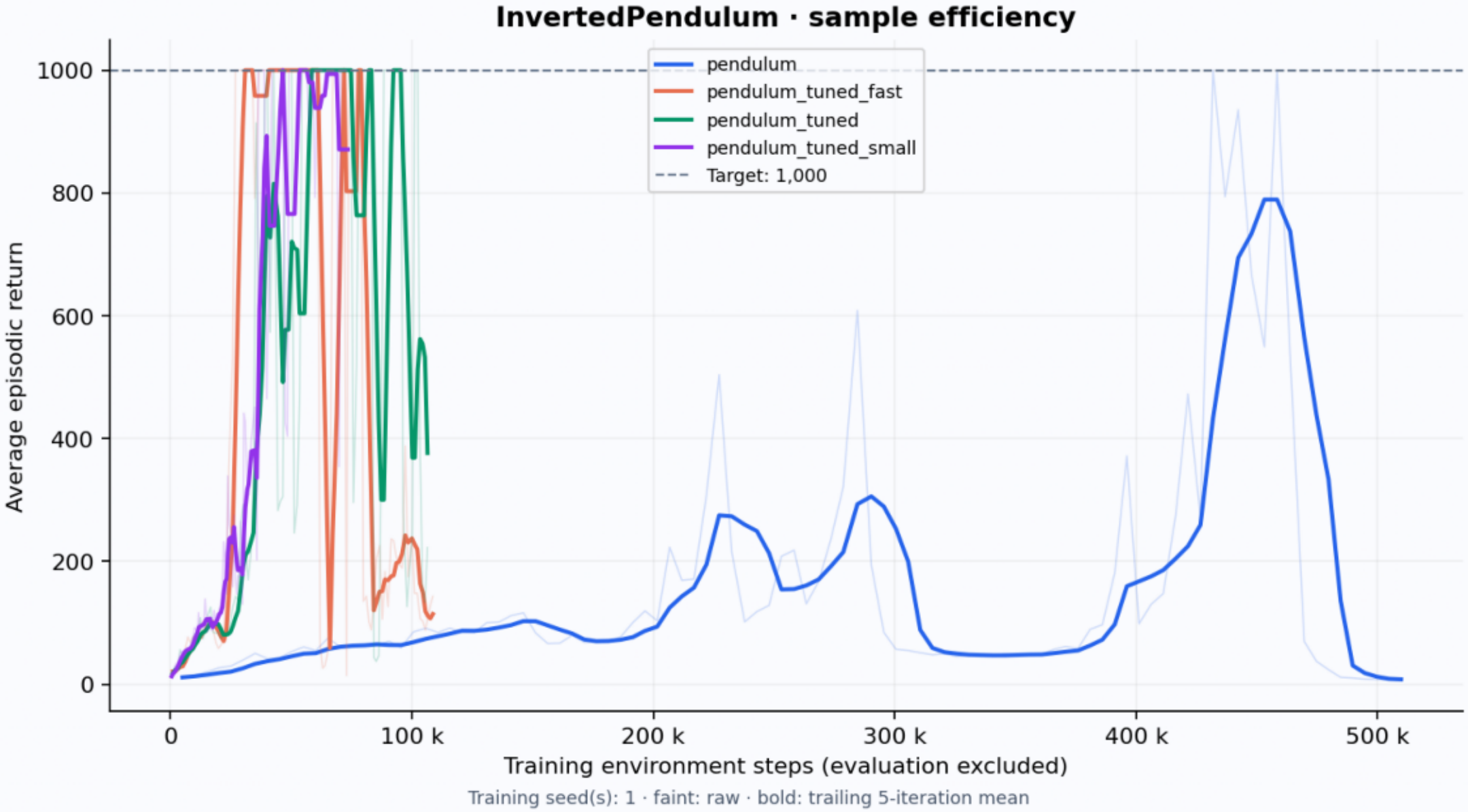

CartPole · 1,000-step training batches
Average episodic return
200
175
150
125
100
75
50
25
0
20 k
40 k
60 k
80 k
100 k
Training environment steps (evaluation excluded)
cartpole_na
cartpole_rtg_na
cartpole_rtg
cartpole
Target: 200
Training seed(s): 1 · faint: raw · bold: trailing 5-iteration mean


CartPole · 4,000-step training batches
Average episodic return
200
175
150
125
100
75
50
25
0
50 k
100 k
150 k
200 k
250 k
300 k
350 k
400 k
Training environment steps (evaluation excluded)
cartpole_lb_na
cartpole_lb_rtg_na
cartpole_lb_rtg
cartpole_lb
Target: 200
Training seed(s): 1 · faint: raw · bold: trailing 5-iteration mean


HalfCheetah · value baseline comparison
Average episodic return
500
250
0
−250
−500
−750
−1000
−1250
0
100 k
200 k
300 k
400 k
500 k
Training environment steps (evaluation excluded)
cheetah_baseline_bgs1 (seed 1)
cheetah_baseline_na (seed 1)
cheetah_baseline (seed 1)
cheetah_baseline (seed 2)
cheetah_baseline (seed 3)
cheetah_baseline_torch (seed 1)
cheetah (seed 1)
Target: 300
Training seed(s): 1, 2, 3 · faint: raw · bold: trailing 5-iteration mean


HalfCheetah · fitting the value baseline
cheetah_baseline_bgs1 (seed 1)
cheetah_baseline_na (seed 1)
cheetah_baseline (seed 1)
cheetah_baseline (seed 2)
cheetah_baseline (seed 3)
cheetah_baseline_torch (seed 1)
Training-batch baseline MSE
0
100
200
300
400
500
0
100 k
200 k
300 k
400 k
500 k
Training environment steps (evaluation excluded)
Training seed(s): 1, 2, 3 · faint: raw · bold: trailing 5-iteration mean


LunarLander · GAE bias–variance tradeoff
Average episodic return
200
100
0
−100
−200
−300
−400
0
100 k
200 k
300 k
400 k
Training environment steps (evaluation excluded)
lunar_lander_lambda0.95
lunar_lander_lambda0.98
lunar_lander_lambda0.99
lunar_lander_lambda0
lunar_lander_lambda1
Target: 150
Training seed(s): 1 · faint: raw · bold: trailing 5-iteration mean


InvertedPendulum · sample efficiency
pendulum
pendulum_tuned_fast
pendulum_tuned
pendulum_tuned_small
Target: 1,000
Average episodic return
0
200
400
600
800
1000
0
100 k
200 k
300 k
400 k
500 k
Training environment steps (evaluation excluded)
Training seed(s): 1 · faint: raw · bold: trailing 5-iteration mean

Exact experiment commands

Generated exact benchmark commands; run from the repository root.
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 MKL_NUM_THREADS=1
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 1000 --exp_name cartpole --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 1000 -rtg --exp_name cartpole_rtg --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 1000 -na --exp_name cartpole_na --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 1000 -rtg -na --exp_name cartpole_rtg_na
--no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 4000 --exp_name cartpole_lb --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 4000 -rtg --exp_name cartpole_lb_rtg
--no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 4000 -na --exp_name cartpole_lb_na --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name CartPole-v0 -n 100 -b 4000 -rtg -na --exp_name cartpole_lb_rtg_na
--no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name HalfCheetah-v4 -n 100 -b 5000 -eb 3000 -rtg --discount 0.95 -lr
0.01 --exp_name cheetah --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name HalfCheetah-v4 -n 100 -b 5000 -eb 3000 -rtg --discount 0.95 -lr
0.01 --use_baseline -blr 0.01 -bgs 5 --exp_name cheetah_baseline --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name HalfCheetah-v4 -n 100 -b 5000 -eb 3000 -rtg --discount 0.95 -lr
0.01 --use_baseline -blr 0.01 -bgs 1 --exp_name cheetah_baseline_bgs1 --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name LunarLander-v2 --ep_len 1000 --discount 0.99 -n 200 -b 2000 -eb
2000 -l 3 -s 128 -lr 0.001 --use_reward_to_go --use_baseline --gae_lambda 0 --prune_lunar_particles
--exp_name lunar_lander_lambda0 --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name LunarLander-v2 --ep_len 1000 --discount 0.99 -n 200 -b 2000 -eb
2000 -l 3 -s 128 -lr 0.001 --use_reward_to_go --use_baseline --gae_lambda 0.95 --prune_lunar_particles
--exp_name lunar_lander_lambda0.95 --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name LunarLander-v2 --ep_len 1000 --discount 0.99 -n 200 -b 2000 -eb
2000 -l 3 -s 128 -lr 0.001 --use_reward_to_go --use_baseline --gae_lambda 0.98 --prune_lunar_particles
--exp_name lunar_lander_lambda0.98 --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name LunarLander-v2 --ep_len 1000 --discount 0.99 -n 200 -b 2000 -eb
2000 -l 3 -s 128 -lr 0.001 --use_reward_to_go --use_baseline --gae_lambda 0.99 --prune_lunar_particles
--exp_name lunar_lander_lambda0.99 --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name LunarLander-v2 --ep_len 1000 --discount 0.99 -n 200 -b 2000 -eb
2000 -l 3 -s 128 -lr 0.001 --use_reward_to_go --use_baseline --gae_lambda 1 --prune_lunar_particles
--exp_name lunar_lander_lambda1 --no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name InvertedPendulum-v4 -n 100 -b 5000 -eb 1000 --exp_name pendulum
--no_gpu --rollout_backend numpy
uv run src/scripts/run.py --env_name InvertedPendulum-v4 -n 100 -b 1000 -eb 1000 -rtg -na --discount
0.99 -lr 0.01 --use_baseline --gae_lambda 0.95 -bgs 10 --exp_name pendulum_tuned --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name InvertedPendulum-v4 -n 100 -b 500 -eb 1000 -rtg -na --discount
0.99 -lr 0.005 --use_baseline --gae_lambda 0.98 -bgs 10 --exp_name pendulum_tuned_small --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name InvertedPendulum-v4 -n 100 -b 1000 -eb 1000 -rtg -na --discount
0.99 -lr 0.02 --use_baseline --gae_lambda 0.95 -bgs 10 --exp_name pendulum_tuned_fast --no_gpu
--rollout_backend numpy
uv run src/scripts/run.py --env_name HalfCheetah-v4 -n 100 -b 5000 -eb 3000 -rtg --discount 0.95 -lr
0.01 --use_baseline -blr 0.01 -bgs 5 --exp_name cheetah_baseline --no_gpu --rollout_backend numpy
--seed 2
uv run src/scripts/run.py --env_name HalfCheetah-v4 -n 100 -b 5000 -eb 3000 -rtg --discount 0.95 -lr
0.01 --use_baseline -blr 0.01 -bgs 5 --exp_name cheetah_baseline --no_gpu --rollout_backend numpy
--seed 3

uv run src/scripts/run.py --env_name HalfCheetah-v4 -n 100 -b 5000 -eb 3000 -rtg --discount 0.95 -lr 0.01 --use_baseline -blr 0.01 -bgs 5 --exp_name cheetah_baseline_na --no_gpu --rollout_backend numpy -na
uv run src/scripts/run.py --env_name HalfCheetah-v4 -n 100 -b 5000 -eb 3000 -rtg --discount 0.95 -lr 0.01 --use_baseline -blr 0.01 -bgs 5 --exp_name cheetah_baseline_torch --no_gpu --rollout_backend torch